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Priority Design in Multi-Target AR-HUD Warning: Evidence from Eye Movement and Behavior of the Novice Driver

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ABSTRACT

Technically, Augmented Reality Head-Up Display (AR-HUD) technology can augment multiple targets in the scene. However, existing literature predominantly focuses on scenarios involving single-target augmentation. The understanding of multi-target augmentation and studies exploring effective presentation methods under such conditions are notably scarce. This study evaluates the efficacy of integrating color-based warning priority design across multi-target scenarios. 45 Participants in different warning modes (Equivalent, Hierarchical, and Baseline) view AR-augmented driving videos and respond to risky targets. Their behavioral performance and eye-tracking data are compared. Findings indicate that the equivalent warning mode, lacking in priority design, adversely affects driver performance, prolongs reaction times, and elevates saccade counts, and gaze entropy. Conversely, the hierarchical warning mode significantly ameliorates driver reaction times and the time to first fixation, while also reducing saccade counts and gaze entropy, demonstrating the efficacy of the warning priority design. The findings provide insight into the design of AR-HUD with multi-target augmentation.

KEYWORDS

AR-HUD; multi-target augmentation; target number; warning priority design

1. Introduction

Vehicles equipped with Augmented Reality Head-up Display (AR-HUD) technology are increasingly entering the market (Coleman Merenda et al., 2018). AR-HUD is an advanced driving assistance system that integrates real and virtual elements within the driver's line of sight (Karatas et al., 2020), facilitating the direct perception of artificial information superimposed on physical reality (Kim et al., 2016). A key advantage of AR-HUD is its ability to align virtual warnings with the direction and location of hazards (Yontem et al., 2021), communicate spatially relevant information about potential dangers (Currano et al., 2021), direct drivers' attention to critical situations (Kim et al., 2016; 2018), and enhance safety by avoiding potential risks without diverting the driver's gaze from the road (Herbert et al., 2017; Minh Tien et al., 2016). Studies have demonstrated the effectiveness of AR warnings, such as improving situational and spatial awareness (e.g., Kim & Hwang, 2016; Ng-Thow-Hing et al., 2013) and reducing response times (e.g., Wang et al., 2023), through visual cues.

However, most of these studies focus on scenarios with a single risky target (Wu et al., 2022), while more complex situations involving multiple road users and objects (i.e., multi-target scenarios) are rarely addressed. In reality, driving often involves a multitude of objects in the scene (Faure et al., 2016), and the typical semantic segmentation used in

Advanced Driver Assistance Systems (ADAS) assumes that all objects are equally important and aims to segment them accurately (Chen et al., 2019). Based on existing detection algorithms, such as Deep Convolutional Neural Networks (DCNNs), AR-HUD systems would augment all target information detected by vehicle sensors and analytical systems (Currano et al., 2021). Current prototype interfaces for in-vehicle AR-HUDs also predominantly utilize multi-target augmentation. It is conceivable that future AR-HUD interfaces will augment multiple targets in the driving environment, including pedestrians, vehicles, traffic lights, and street signs within a certain range, rather than focusing solely on risky targets.

Compared to single-target augmentation, multi-target augmentation could have a significantly different impact on drivers' visual patterns and situational awareness. While it may aid in focusing attention on relevant driving-related targets, it could also distract drivers and potentially diminish the effectiveness of AR warnings. In particular, novice drivers often have difficulty scanning the area ahead effectively (Underwood et al., 2003) and have limited attentional resources (van Leeuwen et al., 2015), so multi-target augmentation may have a more pronounced effect on them. It is crucial to explore the potential trade-offs of multi-target augmentation and to investigate how to effectively present

AR warnings that assist these drivers in identifying risky targets amidst a multi-target environment.

1.1. Multi-target augmentation in AR-HUD and potential impacts

The integration of AR-HUD with multi-target augmentation may have a profound influence on drivers' visual attention and situational awareness. On the one side, the multi-target augmentation in AR-HUD can, to a certain extent, help drivers allocate their visual attention to recognize numerous targets in the scene. Specifically, driving is inherently a complex activity, necessitating the perception of various traffic targets (Chung et al., 2022; Herbert et al., 2017). Pedestrians, vehicles, and traffic signals (e.g., road signs and traffic lights), are crucial targets for aiding in the comprehension of road conditions (Chen et al., 2019; Herbert et al., 2017). A primary task for drivers is to detect, recognize, and allocate attention to these pertinent targets (Arexis et al., 2017; Colley et al., 2021). Neglecting to do so can lead to oversight of route details or even collisions with other vehicles or pedestrians (Luo et al., 2008). Scholars have proposed that the user interface of Advanced Driver Assistance Systems (ADAS) should alert drivers to significant signs and obstacles by sensing external scene targets (Kastner et al., 2011; Wu et al., 2022). Consequently, the integration of AR-HUD with object annotation technology serves to enhance the visual perception of most targets within a specific area (Cavanagh & Alvarez, 2005; Ohn-Bar & Trivedi, 2017). By emphasizing relevant objects and guiding the gaze in a manner that respects the process of information acquisition (Eyraud et al., 2015), this approach facilitates drivers' perception of scene elements. Prior research has already prototyped an AR-HUD interface that highlights traffic objects like vehicles, pedestrians, and road signs, confirming its efficacy in enhancing situational awareness (Colley et al., 2021).

However, multi-target augmentation in AR-HUD interfaces could also negatively impact visual attention and situational awareness. Human attention capacity is inherently limited (Lee et al., 2024), and it is not feasible to attend to every simultaneous target (Broadbent, 1957; Sanocki et al., 2015). Multiple targets may compete for attention, increase the cognitive load (Merenda et al., 2018; Zhang & Lee, 2021), and hinder the allocation of attention to critical targets (Currano et al., 2021; Lee et al., 2024). Recognizing critical objects surrounded by numerous objects can be challenging (Albonico et al., 2018; Chaney et al., 2014; Levi, 2008). Traditional psychological studies have shown that the presence of nearby objects affects the identification of critical objects (Xia et al., 2020), and the perception of a target can be significantly compromised in the presence of similar flanking elements (Chicherov & Herzog, 2015; Levi, 2008; Pichlmeier & Pfeiffer, 2021; Sanocki et al., 2015). When employing multiple AR graphics to augment various targets in the driving environment, such as pedestrians, vehicles, traffic signals, and road signs, these graphics act as multiple information sources. While AR-HUD is designed to capture attention, an increase in the number of augmented targets

could exacerbate attention competition. For instance, Muela et al. (2021), using two bounding boxes to simultaneously augment two types of hazards (i.e., potential and developing hazards), found that the presence of dual targets significantly impaired drivers' performance. Similarly, Kim et al. (2016) augmented all detected pedestrians within the sensor's field and observed that drivers needed to exert additional effort to discern dangerous pedestrians among cluttered bounding boxes. It has been suggested that an overload of information in the line of sight could impair driving performance (Tretten et al., 2009). In particular, there is a consistent discrepancy in distance between the projected AR graphics and real-world objects when utilizing an AR-HUD. The distance of AR graphics remains constant, whereas real-world objects are situated at varying distances. This discrepancy can impact the eye's optical power, leading to an accommodation-convergence conflict (Kang et al., 2019; Ng-Thow-Hing et al., 2013). Prior research indicates that accommodation-convergence conflict correlates with visual fatigue (Shibata et al., 2011) and diminished attentiveness (Kang et al., 2019). For instance, Kang et al. (2019) demonstrated that the distance disparity between AR images and real objects affects the accuracy of object recognition. Consequently, we hypothesize that when viewing multiple AR-augmented graphics and external objects simultaneously, a driver's visual attention may be more significantly compromised than when a single target is augmented, potentially increasing the difficulty in recognizing critical targets.

Overall, while some studies have examined the impact of multiple AR graphics on visual attention and shown that an increase in AR graphics can lead to perceived visual complexity (Currano et al., 2021; Lee et al., 2024), they have largely focused on static screen-fixed graphics. It has been suggested that dynamic stimuli may result in greater performance decrements than static stimuli in visual vigilance tasks (Koelega et al., 1990). Given the dynamic and world-fixed graphical nature of multi-target augmentation, these dynamic AR graphics may have more pronounced effects on visual attention than static AR graphics. Furthermore, in dynamic driving contexts, the application of multi-target augmentation, which includes pedestrians, vehicles, road signs, and traffic lights, may impact drivers' perception of the significance of these environmental elements and their overall situational awareness. Particularly, when environmental cues are ambiguous, individuals are required to perform diagnostic procedures to ascertain the underlying issues (Orasanu et al., 1993). The potential blurring effect of such augmentation on situational awareness warrants further investigation.

1.2. Priority design in multi-target augmentation condition

The preceding discussion suggests that multi-target augmentation in AR-HUD can be a double-edged sword, particularly for novice drivers. Novice drivers often struggle with identifying traffic targets (Koelega et al., 1990; van Leeuwen et al., 2015), making multi-target augmentation a potentially

essential aid. However, these drivers also have limited spare attentional resources (Duncan et al., 1991; van Leeuwen et al., 2015), which may exacerbate the visual attention competition effects associated with multi-target augmentation. Moreover, to a certain extent, the similarity in multi-target augmentation can blur the significance of these pieces of information. Without discerning the significance of various pieces of information, individuals might erroneously believe they are focusing on the most critical data (Endsley & Endsley, 1995). In particular, novices may acquire less information due to their inability to differentiate between similar inputs (Endsley & Endsley, 1995). Therefore, while sufficient AR warnings were present in this condition, the difficulty in determining their relevance to important goals ultimately makes it difficult to focus on the most important one promptly as well. Since situational awareness encompasses a comprehensive understanding of all elements (Endsley & Endsley, 1995; Teichmann & Motus, 2021), this could affect drivers' comprehension of the dynamic driving environment. Improperly arranged additional information could overshadow its potential benefits, especially in interfaces that must support time-critical tasks (Beck et al., 2010). To ensure the effectiveness of AR-HUD for the majority of drivers, including novice drivers, we focus on exploring an effective interface design from the perspective of multi-target augmentation. We propose integrating the concept of warning priority with multi-augmentation.

The Intelligent Transport Systems (ITS) of the United Nations Economic Commission for Europe (UNECE) has clearly defined the concept of warning priority as the establishment of relative importance among multiple pieces of information and their ranking in the presentation emphasis. It is undeniable that not all targets in the driving environment are equally important (Chen et al., 2019; Ohn-Bar & Trivedi, 2017). Wu et al. (2022) has delineated the importance levels of driving-related targets based on the risk they pose to driving, advocating for a high importance level for targets that present a direct crash risk or induce serious traffic violations, and a medium importance level for those that influence future driving decisions but do not pose an immediate crash risk. For instance, a road sign indicating a turn is considered a medium-importance target; it may necessitate a prompt response from the driver, but missing this signal does not necessarily result in dangerous consequences. Multi-target augmentation involves world-fixed graphics, which closely tie the nature of the warnings to the nature of the targets themselves, facilitating the establishment of warning priorities in the AR-HUD interface. Prior research has suggested that advanced driver assistance systems should provide drivers with alerts corresponding to their importance (risk) levels, focusing more on risky targets closely related to safety (Chen et al., 2019) to ensure that information is presented within the bounds of available attention resources (Herbert et al., 2017; Pankok & Kaber, 2018). We posit that a warning prioritization design, predicated on target risk levels, can optimize the utilization of world-fixed graphics in AR-HUDs. Such a design can enhance drivers' focus on critical driving information, particularly during

emergencies, by facilitating rapid identification of risky targets and thereby mitigating the visual attention competition induced by multiple targets. Concurrently, discerning critical environmental cues is imperative for achieving a high level of situational awareness (Endsley & Endsley, 1995). From a design standpoint, the rational interpretation of presented elements to clarify what requires the operator's attention and understanding is fundamental to enhancing situational awareness (Endsley & Endsley, 1995). In this context, a warning priority design that categorizes the importance of warning graphics can aid drivers in comprehending the risk levels among various warnings, preventing confusion over ambiguous importance levels, and fostering situational awareness, which is vital in the complex and dynamic driving environment.

Furthermore, concerning the aspect of the implementation of warning priority, the ITS has advocated that high-priority warnings can be distinguished from others by altering visual characteristics. For example, This recommendation is supported by conventional multi-target tracking studies, which have found that a significant difference in the description between the critical target and non-critical targets leads to efficient search (Herbert et al., 2017), and that stimuli with high visual saliency are prioritized for attentional selection in cluttered scenes (Herbert et al., 2017; Jung et al., 2020). It has been reported that visual separation between critical targets and other elements, facilitated by surface features or semantic information, can enhance tracking performance (Kamkar et al., 2020; Wei et al., 2016). As such, color could be an effective means as it can differentiate visual saliency and convey different semantic meanings. Some researchers have suggested that the most critical target could be presented in a salient color to reduce the detrimental effects of clutter, as color is a crucial factor in determining visual search efficiency (Beck et al., 2010; Breslow et al., 2009; Rosenholtz et al., 2007). It is logical to use different colors to represent multiple targets based on their importance (risk level) to the driving task. The color contrast would aid drivers in identifying and quickly focusing on the risky target among multiple AR warnings.

However, AR-HUD systems may encounter hue-shifting issues due to ambient brightness, background color, and other factors (Gabbard et al., 2022; Merenda et al., 2016). The dynamic nature of AR graphics might also impact the effectiveness of detecting warnings based on color differentiation. Therefore, while previous researchers have attempted to use different colored AR bounding boxes for cueing different types of targets (Muela et al., 2021), the effectiveness of warning priority, differentiated by color, on visual allocation still requires validation.

1.3. Motivation

In summary, as previous research on AR-HUD has predominantly focused on single-target augmentation scenarios, this study endeavors to explore the significance of multi-target augmentation and assess the efficacy of prioritization warnings differentiated by color within this context. Especially

for novice drivers, who frequently face difficulties in effectively scanning the forward area (Underwood et al., 2003), tend to focus their attention within a limited scope (Chapman et al., 2002), and are inclined to exhibit error-prone search behaviors (Twisk, 1995), there is an urgent need for real-time multi-target augmentation. However, their constrained attentional resources (Kerruish et al., 2022), inexperience (Xiang et al., 2024), and incapacity to accurately evaluate potential risks (Twisk, 1995) render them more vulnerable to the impact of multiple AR warnings. Therefore, this study also aims to provide valuable insights from the perspective of novice drivers for the enhancement of future AR-HUD interface designs.

Additionally, while most existing research has concentrated on behavioral metrics, there is a dearth of studies examining the influence of AR-HUD interfaces on eye movement measures. This study aims to delve into visual attention using a broader range of eye-movement metrics, encompassing time to first fixation, saccade count, and gaze entropy (both stationary gaze entropy and gaze transition entropy). These entropy metrics provide direct assessments of the overall distribution of eye movements and the predictability of spatial transitions (Krejtz et al., 2015), thereby reflecting the complexity of eye movements (Schieber & Gilland, 2008) and the visual search strategies employed by participants (van Maarseveen et al., 2018). By integrating gaze entropy with conventional eye-tracking metrics, this study aims to augment the findings of previous eye movement studies and enable a quantitative analysis of variations in eye movement patterns across different warning modalities.

2. Method

2.1. Participants

Due to the limitations in G Power's ability to handle mixed-design analyses, an a priori power analysis was conducted using MorePower 6 (Campbell & Thompson, 2012). The analysis was set with a significance level of 0.05, a power of 0.8, and an effect size of 0.2. The findings suggested that a total sample size exceeding 42 is necessary to achieve the desired statistical power. Consequently, this study included 45 participants (23 males and 22 females), all of whom were students from Zhejiang Sci-Tech University in Hangzhou, China. The average age of the participants was 22.7 years ($SD = 2.76$). All participants reported having normal vision or vision that was corrected to normal, and they all possessed a valid Chinese driver's license. The average driving experience was 1065 kilometers and 2.7 years, respectively. Written informed consent was obtained from each participant, following the approved research protocol of the Institutional Review Board at the Department of Psychology, Zhejiang Sci-Tech University. Upon completion of the experiment, all participants were compensated.

2.2. Experimental equipment

The experimental environment, as referred to Wang et al. (2023), comprised a 72-inch high-definition liquid crystal

display (LCD) screen (Huayin Vision Technology Co., CN), a desktop eye-tracking device (SMI iView Red-M, SensoMotoric Instruments, Berlin, Germany) and a chin support. This monitor, featuring a resolution of 1920 by 1080 pixels, served as the medium for presenting driving video clips. The eye-tracking device was capable of calibration at nine points, ensuring recordings had a spatial error of less than 0.8 deg in both x and y directions. It operates at a sampling rate of 120 Hz with a latency of less than 20 milliseconds. The systematic error is 0.5 and the spatial resolution is 0.1°. The screen was positioned at a vertical distance of 1.7 meters from the participants, equating to a visual angle of 41.64 by 24.15 degrees. The eye tracker was situated on a table, 65 centimeters distant from the participants. A chin support was used to stabilize participants' heads to eliminate movement (Wang et al., 2016). Furthermore, a keyboard linked to the host computer facilitated the collection of behavioral response times. The configuration of the experimental setup is depicted in Figure 1. In addition, the scenarios depicted in the video clips were constructed using a driving simulator, which incorporated driving simulator software (STISIMDRIVE M100K, Systems Technology, Inc., Hawthorne, CA, USA), a steering wheel, an accelerator, and a brake pedal (Logitech MOMO, Newark, CA, USA). The driving simulation software was used to build the driving scenarios, while the steering wheel, accelerator, and brake pedals were used to control the ego-vehicle's movement within the scenario when recording videos.

2.3. Materials

Previous research has demonstrated the efficacy of brief video clips in assessing situation awareness in driving studies (Currano et al., 2021) and in examining drivers' hazard perception (Crundall et al., 2012). Crundall and Underwood (2011) have highlighted the advantage of this method, wherein each participant is exposed to identical traffic conditions. To maintain experimental control, we elected to emulate prior studies by utilizing video clips as the experimental stimuli. We initially constructed driving scenarios using medium-fidelity driving simulation software, which facilitated a first-person perspective and a dynamic driving environment. Subsequently, an operator maneuvered a vehicle through a pre-constructed driving scenario and recorded the corresponding driving footage using the software. During the drive, the operator controlled the vehicle's movements through the steering wheel, accelerator, brake pedal, and other devices within the control system, executing operations such as acceleration and deceleration at appropriate moments. This procedure was primarily designed to provide a more authentic and immersive driving experience. A total of 20 videos were developed, each depicting a six-lane highway with traffic flowing in both directions. Vehicles, pedestrians, traffic lights, and road signs, including directional and speed limit indicators, were integrated into the highway scenes, with all elements appearing at random intervals. In each video, the vehicle is shown navigating the highway, encountering other vehicles, pedestrians, road signs, and traffic lights.

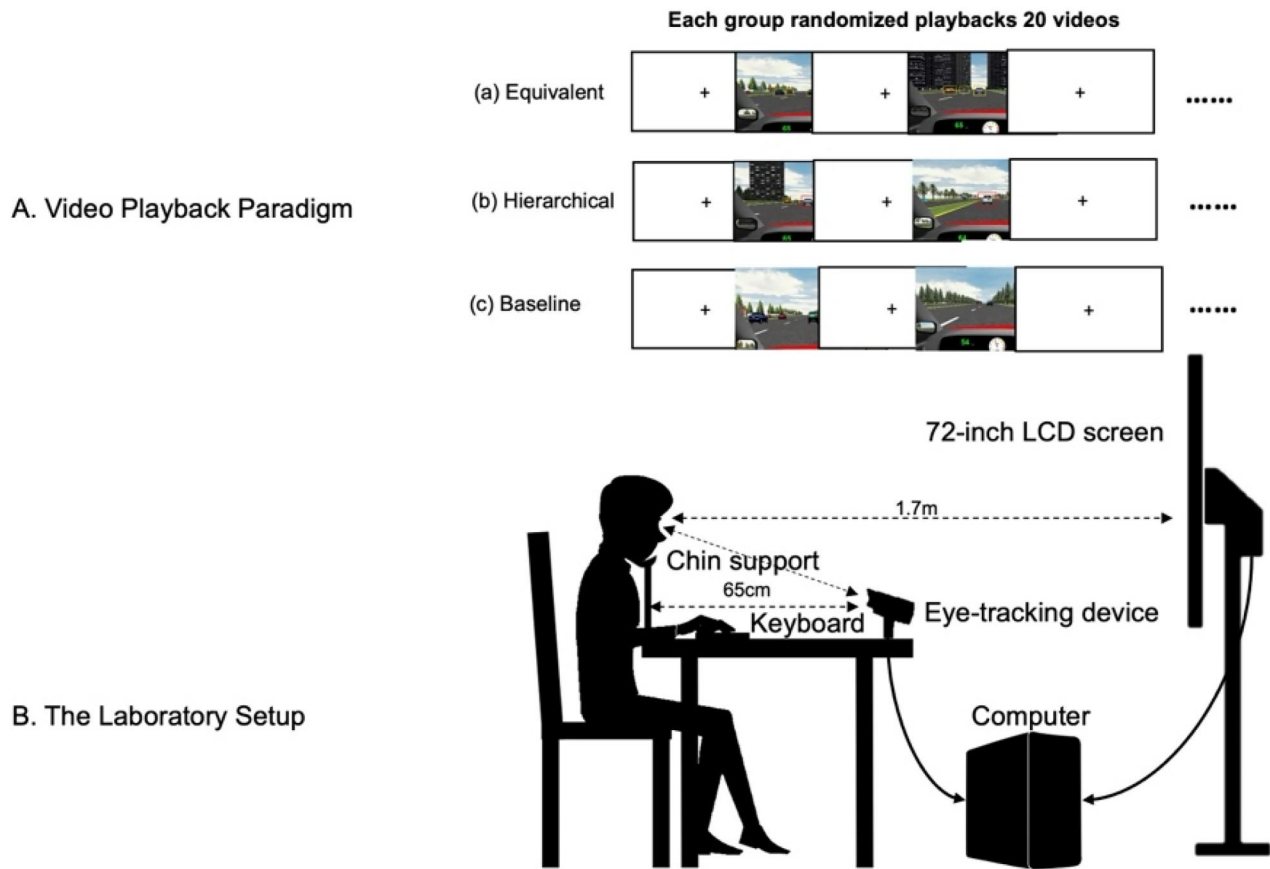


Figure 1. Illustration of video playback example and demonstration of the laboratory setup.

Drawing from the study by Cheng et al. (2023), we incorporated two types of risk targets: a leading vehicle that suddenly brakes and a pedestrian that suddenly crosses the road. To explore the potential impact of varying multi-augmentation conditions (e.g., five or two targets), on the detection of the risky target, we manipulated the number of concurrent targets within the scenario. This was achieved by controlling the number of additional targets present simultaneously with the risky target. We selected traffic lights, road signs, other vehicles, or pedestrians at random to serve as these additional targets. We devised four distinct risky target event scenarios: (1) a scenario with five targets featuring a braking leading vehicle as the risky target, (2) a scenario with five targets featuring a crossing pedestrian as the risky target, (3) a scenario with two targets featuring a braking leading vehicle as the risky target, and (4) a scenario with two targets featuring a crossing pedestrian as the risky target. To enhance the realism of the driving scenarios, we also included other unpredictable events, such as the vehicle speeding or running a red light. Each video randomly featured 0 to 3 of these unpredictable events, including pedestrian collisions, rear-end collisions, speeding, and red-light violations, with the timing of these events determined randomly. Following the methodology of Wang et al. (2023), each video was set to a duration of 30 seconds.

Subsequently, we introduced different AR graphics into the 20 videos to create two distinct AR warning modes: Equivalent warning mode and Hierarchical warning mode.

Equivalent warning modes represent the case of no prioritization design, and hierarchical warning modes represent the case of prioritization design. Given that both yellow and red are appropriate for conveying warnings (Friedrich & Vollrath, 2022), and red signifies higher urgency (Friedrich & Vollrath, 2022), yellow was designated as the warning color for the foundational multi-target augmentation (equivalent warning mode), while red or yellow was employed based on the target's risk level in the hierarchical warning mode. Both colors were specified in RGB format, with red defined as [255, 0, 0] and yellow as [255, 255, 0]. Bounding box graphics, known for their effectiveness in AR warnings, were utilized in this experiment. In the experiment, we referred to the definition by Wu et al. (2022), categorizing objects within a certain range of the ego-vehicle as medium importance level, while objects within the same range that pose collision risks are classified as high importance level. Since time headway is an important variable in the distance-keeping algorithms in driving assistant system (Siebert et al., 2014) and the National Road Administration of Sweden recommends maintaining a time headway of three seconds (Vogel, 2003), we defined a three-second time headway on the longitudinal distance as the range threshold to trigger the AR warning. To be more specific, we initially employed a time headway threshold to trigger AR warnings. An AR warning is immediately activated when an object's time headway relative to the ego-vehicle is less than or equal to three seconds. In the equivalent warning mode, all objects

meeting the trigger conditions are marked with a yellow AR bounding box, without distinguishing between those that pose collision risks. Building on this, in the hierarchical warning mode, the color of the AR bounding box is determined based on both the object's time headway and collision risk. If an object is within 3 seconds of the vehicle but does not pose a collision risk during this period, the system uses a yellow AR bounding box to alert the driver to pay attention without necessitating immediate action. Conversely, if an object is approximately 3 seconds away from the vehicle and poses a collision risk during this period, the system employs a red AR bounding box to warn the driver to take action. During this process, leveraging the adjustable nature of the simulated scenario, to ensure the reasonable implementation of the warning system, we pre-controlled the position, distance, and motion parameters of the corresponding objects in the scene, thereby ensuring that only preset crossing pedestrians and suddenly braking vehicles pose collision risks within this duration in the driving scenario. Consequently, the AR bounding box is designed to highlight targets approaching the vehicle within roughly three seconds, with its size adjusted according to the dimensions of the target in the video. The unaltered video clips served as the baseline for comparative analysis. Figures 2 and 3 illustrate examples of the different warning modes applied to the two types of risky targets in scenarios with five or two target numbers.

2.4. Experimental design

A 3×2 mixed design was manipulated in the present experiment. The independent variables comprised the warning mode (with levels: Equivalent warning, Hierarchical

warning, and Baseline) as a between-subjects factor, and the target number (with levels: five targets and two targets) as a within-subjects factor.

The warning mode was operationalized across three distinct conditions: equivalent warning, hierarchical warning, and baseline. In the equivalent warning condition, a uniform yellow AR bounding box was employed to alert participants to targets in the scene, irrespective of their risk level, primarily to examine the influence of multi-target augmentation. Conversely, the hierarchical warning condition utilized varying colors to distinguish warnings based on the targets' risk levels, primarily to assess the efficacy of warning prioritization. The baseline condition was devoid of any AR warnings, serving as a control for comparison.

The target number refers to the total count of targets present in the scenario at the time the risky target emerges. Prior research studies of multiple object tracking (MOT) suggest that individuals can track up to four to five identical targets simultaneously (Beck et al., 2010; Cavanagh & Alvarez, 2005; Hyönä et al., 2019), but participants exhibit higher rates of perceptual error and higher response latency when the number of tracked objects is five (Pylyshyn & Storm, 1988). Furthermore, when the tracking targets are changed to targets with distinct features in studies of multiple identity tracking (MIT), it suggests that an individual's ability to track simultaneously may be limited to two different targets (Horowitz et al., 2007; Hyönä et al., 2019). A prior study confirmed that when it comes to tracking the features and location of targets, participants can track the number of targets of approximately 1.6 items (Hyönä et al., 2019). Given that the main thrust of the study was to explore the effects of multiple target augmentation, we combine the results of the MOT and MIT studies, as well as the

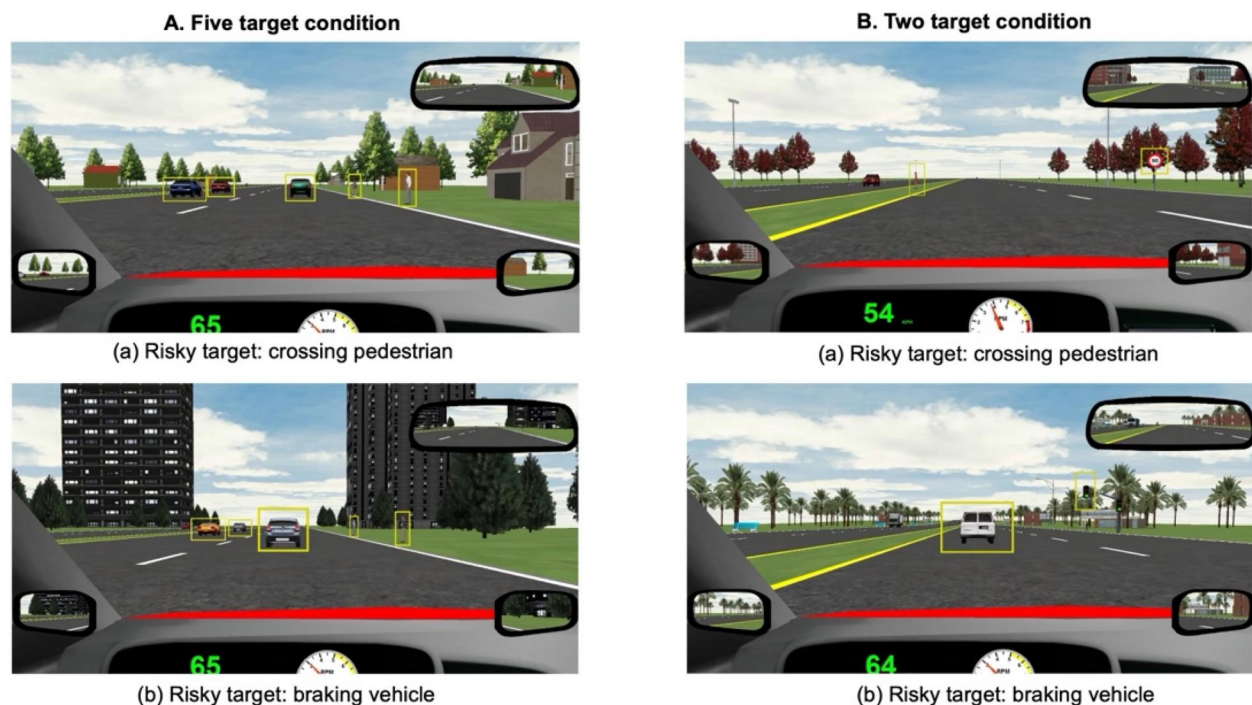


Figure 2. The equivalent warning modes were examined in the driving simulator study to investigate the impact of multi-target augmentation. *Note:* We created four risky scenarios based on target numbers and risky target types (i.e., risky target events).

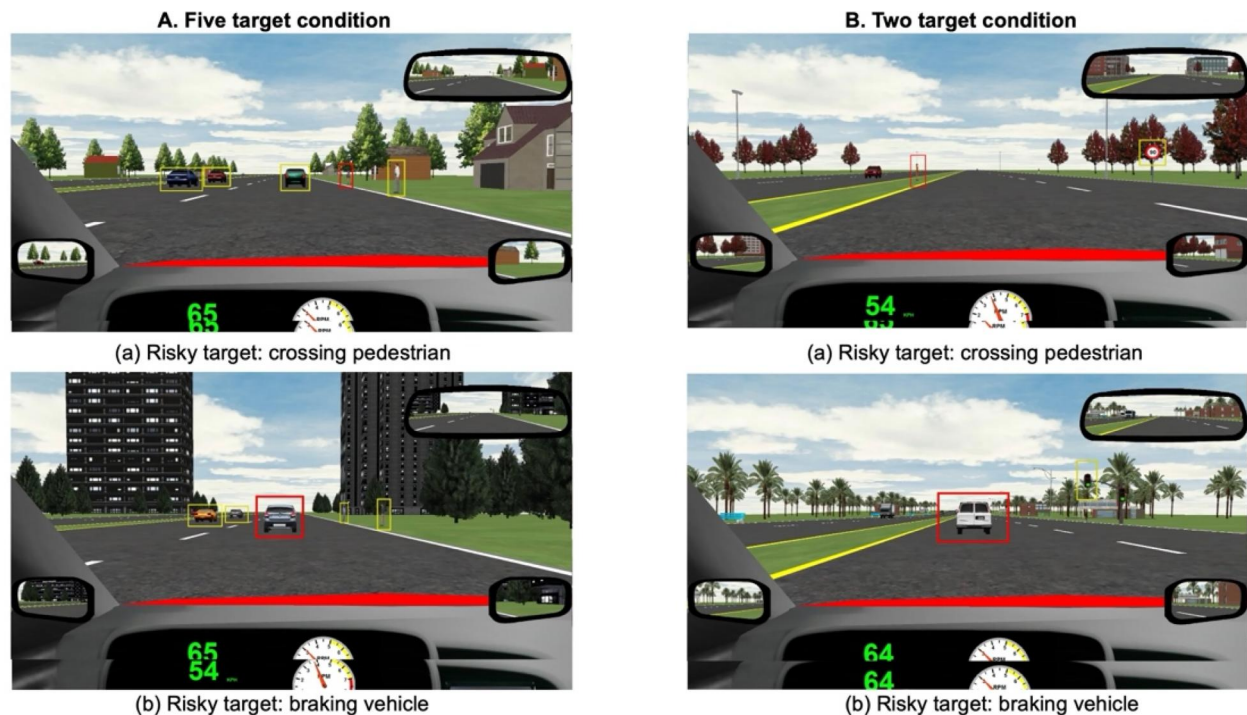


Figure 3. The hierarchical warning modes were examined in the driving simulator study.

complexity and dynamics of the driving scenario, and thereby suspect two or five targets presented simultaneously in the driving scenario can serve as a key entry point for target numbers. It can realistically reflect an individual's visual attention when confronted with multiple targets. Moreover, a previous study examining the effects of AR graphic numbers on vision similarly used two and five AR graphic numbers as extreme levels as well (Lee et al., 2024). Eventually, in the experiment, the target number in the risky driving scenario was set to two levels: two targets and five targets, respectively.

The dependent variables encompassed both behavioral performance and eye-tracking data. Behavioral performance metrics included response rate and reaction time. However, the response rate was omitted from the results due to ceiling effects observed in subsequent analyses. Reaction time was calculated as the interval between the emergence of the risk target and the moment participants provided feedback upon detecting the risk (Cheng et al., 2023). Eye-tracking data analysis focused on time to first fixation (TFF), saccade count, and gaze entropy. Gaze entropy, which includes stationary gaze entropy (H_s) and gaze transition entropy (H_t), is an appropriate measure for dynamic eye movement characteristics (Harezlak et al., 2019). The definitions of eye movement variables are detailed in Table 1.

2.5. Procedure and tasks

Before their initial visit, informed consent was obtained from all participants. Subsequently, participants were assigned to different warning modes by the experimenter, who also familiarized them with the warning system and the tasks involved, and assisted in calibrating their eye movements. Throughout

the formal experiment, participants in all three groups were directed to view the driving video simulations and to immerse themselves in the role of the driver. Participants were instructed to promptly press a button upon detecting an unpredictable event, such as: (1) the vehicle speeding, (2) the vehicle running a red light, (3) a pedestrian crossing the road, or (4) the leading vehicle braking. The inclusion of the first two events was strategic to mitigate the potential for participants to react solely based on anticipatory effects (Wang et al., 2022).

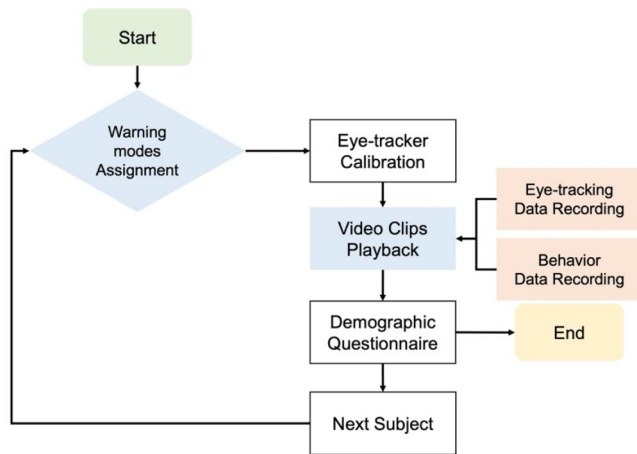
In total, 24 events were distributed randomly across the 20 driving video simulations, with the critical risky target events—namely, the pedestrian crossing and the leading vehicle braking—each occurring four times. During this process, the background program recorded participants' key-stroke reactions in real time, while the desktop eye-tracking device concurrently captured the drivers' scanning behaviors. The participants' simple push-button responses to risky targets served as a metric for their corresponding hazard-response abilities (Crundall et al., 2012). All 20 videos were programmed to play in a randomized sequence. The schematic diagram of the experimental paradigm is depicted in Figure 1(a). Once all videos had concluded, demographic information was collected from participants across the three groups. The detailed procedure is illustrated in Figure 4.

2.6. Eye-tracking data processing

Eye-tracking data is examined with the software package SMI BeGaze3.2.28. Exclusion criteria led to removing three participants from each experimental group due to an eye-tracking ratio that fell below the 80% threshold, leaving a total of 42 participants (22 males, 20 females) for analysis.

Table 1. Definitions of eye-tracking measures.

Metric	Definition
Time to first fixation (TFF)	Time to first fixation (TFF) was defined as the time it took following the onset of the risky target event to first fixate on the risky target (Lai et al., 2013; Vervoort et al., 2013). A low number reflects an early fixation on the risky target, and a higher number represents a later fixation (Crundall et al., 2012).
Saccade count	The total number of saccades counted within the specific duration (i.e., the duration from the appearance of the risky target to the keystroke response) (Lai et al., 2013). The high saccade count reflected an extensive search (Kotval & Goldberg, 1998).
Stationary gaze entropy (H_s)	Stationary gaze entropy is a method in which Shannon's entropy Equation (1) is applied to the probability distribution of fixation coordinates to calculate the average level of uncertainty in the spatial distribution of a sequence of fixations generated in a given timeframe (Shiferaw et al., 2018). $H_s(x) = -\sum_{i=1}^n (p_i) \log_2(p_i) \quad (1)$ Here H_s is entropy for a given set x (time period from the risky target event trigger to participants' key press), i is the number of state spaces of each fixation in x , n is the total number of fixations in x , and p_i is the proportion of fixations landing in a given state space (Goodridge et al., 2024).
Gaze transition entropy (H_t)	Gaze transition entropy, based on the conditional entropy Equation (2) and Markov chain matrices (Shiferaw et al., 2018), is to measure the predictability of visual scanning patterns (Shiferaw et al., 2018). $H_t(x) = -\sum_{i=1}^n p_i \sum_{j=1}^n p_{(i,j)} \log_2 p_{(i,j)} \quad (2)$ Where p_i is as described in stationary gaze entropy, and $p_{(i,j)}$ is the probability of transitioning from state i to state j in one fixation (Devlin et al., 2022). Higher H_t suggests less structured or a more random pattern of scanning behavior (Shiferaw et al., 2018).

**Figure 4.** Experiment procedure.

The primary eye movement metrics collected during this experiment include time to first fixation, saccade count, and gaze entropy. During the eye movement data analysis, to obtain the driver's time to first fixation on the risky target, we first sketch the dynamic area of interest (AOI) related to the risky target according to the position and size of the AR bounding box which cues the risky target in each video. The dynamic AOI is synchronized with the AR warning graphics' dimensions and orientation by adjusting the keyframes within the BeGaze software. Considering the three-second interval from the risky target event's onset to the collision in the video, the dynamic AOI was configured to persist for the same duration. That is, the dynamic AOI is consistent with the AR graphics in each video that suggest risky targets. We then identify the moment the first fixation entered the dynamic AOI and calculate the corresponding time to the first fixation. The standard settings of a minimum duration of 80 milliseconds and a maximal dispersion of 100 pixels were used to define fixations. Besides, a saccade is defined as an eye-movement event occurring between the latest and preceding blink or fixation. We calculate the saccade count in the time period from the onset of the risky target event to the participants' key press. Lastly, to assess

the entropy of the eye movement time series, we collected participants' fixation coordinates in the duration from the emergence of the risky target to the participants' key press. These fixation coordinates were partitioned into a spatial bin of 144 by 90 pixels for the calculation of stationary gaze entropy and gaze transition entropy. The calculation formula of gaze entropy is referred to the Table 1. This strategy is suitable for the formation of state spaces across the visual field, leading to the production of representative transition distributions (Shiferaw et al., 2018).

3. Result

The experimental data were analyzed utilizing IBM SPSS Statistics version 26. A significance level of 0.05 was established for all statistical analyses. Initially, we attempted to compute the results using a repeated measures analysis of variance (ANOVA) with a general linear model (GLM). However, this analysis revealed a bias in the residuals, indicating that the dependent variables did not meet the assumptions necessary for the application of general linear models. Furthermore, given the missing data and the repeated measures, treating individual subjects as random effects was deemed more appropriate. Consequently, we conducted the data analysis using generalized linear mixed models (GLMM), with warning modes and target numbers as fixed effects, and individual subjects as random effects. GLMM integrates the characteristics of linear mixed models, which include random effects, and generalized linear models, which manage non-normal data through the use of link functions and exponential families (Bolker et al., 2009; Lo & Andrews, 2015). It is recognized as the optimal analytical tool for non-normal data involving random effects (Bolker et al., 2009). The effects and interactions of each predictor were assessed by examining the model effects and estimated marginal means. These estimated marginal means were employed to evaluate the impact of warning modes across various conditions, with the Bonferroni correction applied for post-hoc testing.

Table 2. Means and standard deviations (SD) of all variables for different warning modes under different target number conditions.

		Equivalent Warning	Hierarchical Warning	Baseline
Reaction Time (ms)	Five targets	1707.08 (494.87)	1083.10 (454.63)	1533.22 (522.31)
	Two targets	1568.64 (561.90)	983.75 (424.50)	1300.13 (483.86)
Time to First Fixation (ms)	Five targets	1346.24 (744.37)	1050.84 (753.24)	1263.37 (819.18)
	Two targets	998.86 (726.04)	851.18 (770.97)	1053.09 (724.06)
Saccade Count	Five targets	4.87 (2.78)	3.04 (1.84)	3.85 (2.28)
	Two targets	3.98 (2.20)	2.86 (2.16)	3.63 (2.35)
H_s	Five targets	1.92 (0.83)	1.31 (0.81)	1.63 (0.94)
	Two targets	1.69 (0.89)	1.10 (0.91)	1.51 (1.07)
H_t	Five targets	0.30 (0.16)	0.18 (0.19)	0.24 (0.18)
	Two targets	0.26 (0.18)	0.15 (0.19)	0.24 (0.19)

Note: H_s means stationary gaze entropy; H_t means gaze transition entropy.

Table 3. Effects of warning modes and target numbers on all dependent variables.

Dependent variables		<i>F</i>	<i>df</i> ₁	<i>df</i> ₂	<i>p</i>
Behavior Performance	Reaction Time				
	Warning mode	66.732	2	350	<0.001***
	Target number	0.021	1	350	0.885
	Warning mode $\times$ Target number	1.327	2	350	0.267
Visual Performance	Time to First Fixation				
	Warning mode	6.554	2	302	0.002**
	Target number	31.822	1	302	<0.001***
	Warning mode $\times$ Target number	1.991	2	302	0.138
	Saccade Count				
	Warning mode	14.120	2	288	<0.001***
	Target number	1.763	1	288	0.185
	Warning mode $\times$ Target number	1.251	2	288	0.288
	H_s				
	Warning mode	21.067	1	285	<0.001***
	Target number	0.878	1	285	0.350
	Warning mode $\times$ Target number	1.502	2	285	0.225
	H_t				
	Warning mode	21.629	2	285	<0.001***
	Target number	0.568	1	285	0.452
	Warning mode $\times$ Target number	1.868	2	285	0.156

Note: H_s means stationary gaze entropy; H_t means gaze transition entropy.

The mean and standard deviations (SD) for all dependent variables are presented in Table 2, while the effects of warning modes under different target number conditions on all dependent variables are detailed in Table 3.

3.1. Reaction time

The warning mode had a significant main effect on reaction time ($F(2, 350) = 66.732, p < 0.001$). The equivalent warning mode was accompanied by the longest reaction time and was significantly higher than the remaining two modes (Hierarchical: $p < 0.001$; Baseline: $p = 0.002$). The use of the hierarchical warning mode was relatively effective in shortening the reaction time, as evidenced by a significantly shorter reaction time than the remaining modes ($ps < 0.001$). However, We did not find an effect of the target number on the driver's reaction time ($F(1, 350) = 0.021, p = 0.885$).

While there was no significant interaction between the warning mode and target number ($F(2, 350) = 1.327, p = 0.267$), we found that whether there were five or two targets in the scene, the equivalent warning resulted in significantly longer response times than the hierarchical warning ($ps < 0.001$) and baseline (in five-target condition: $p = 0.039$; in two-target condition: $p = 0.012$) respectively. The hierarchical warning was also accompanied by the shortest response times in both conditions ($ps < 0.001$). The detailed results can be found in Figure 5.

3.2. Eye tracking data

3.2.1. Time to first fixation (TTF)

The warning mode had a significant main effect on the time to first fixation ($F(2, 302) = 6.554, p = 0.002$). The use of the hierarchical warning mode can effectively shorten the time to first fixation, resulting in a significantly shorter latency than the equivalent mode ($p = 0.002$) and the baseline ($p = 0.004$). However, the equivalent warning mode did not reach a significant difference with the baseline ($p = 0.796$). There was a significant main effect of the target number on the time to first fixation ($F(1, 302) = 31.822, p < 0.001$). Drivers showed a longer time to first fixation under the five-target condition.

No significant interaction was reported between warning modes and the target number ($F(2, 302) = 1.991, p = 0.138$). The result showed that in the two-target condition, the hierarchical warning resulted in a significantly lower time to first fixation than the equivalent warning ($p = 0.002$) and the baseline ($p = 0.002$).

3.2.2. Saccade count

The warning mode had a significant main effect on the saccade count ($F(2, 288) = 14.120, p < 0.001$). The equivalent warning mode was associated with the highest saccade count, compared to the hierarchical warning mode ($p < 0.001$) and the baseline ($p = 0.033$). And the hierarchical warning mode

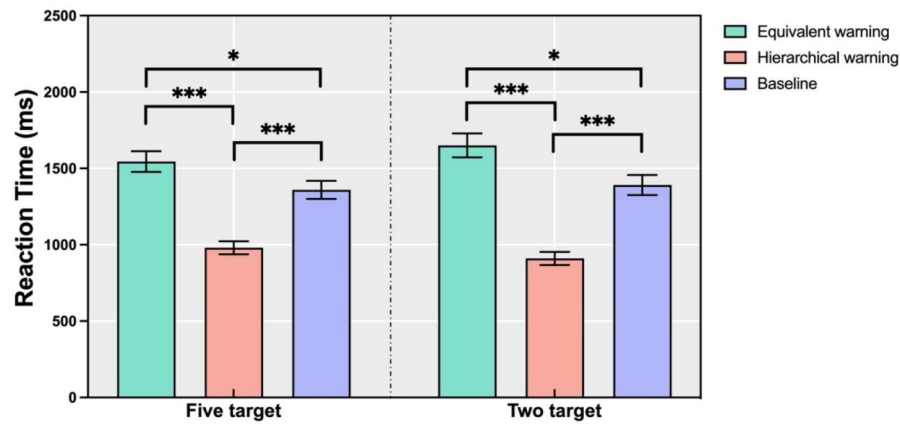


Figure 5. Effects of warning modes on reaction time under different target number conditions. Notes: $+p \leq 0.1$, $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$.

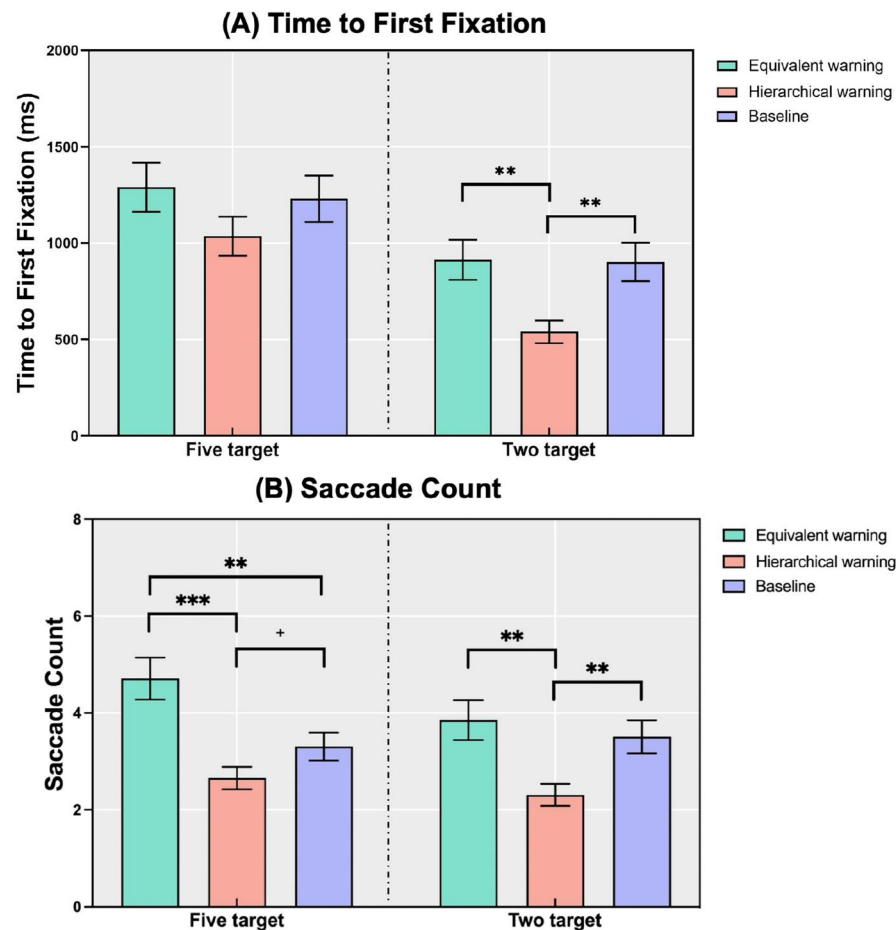


Figure 6. Effects of warning modes on time to first fixation and the saccade count in different target number conditions. Notes: $+p \leq 0.1$, $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$.

achieved a lower saccade count than the baseline ($p = 0.001$). There was no significant main effect of the target number on the saccade count ($F(1, 288) = 1.763$, $p = 0.185$).

No significant interaction was reported between warning modes and the target number ($F(2, 288) = 1.251$, $p = 0.288$). We further analyzed the result and found that in the five-target condition, the equivalent warning was accompanied by a higher saccade count than the hierarchical warning mode ($p < 0.001$) and the baseline ($p = 0.007$), and the driver in the hierarchical warning mode has a marginally lower saccade count than baseline group ($p = 0.080$). In the two-target

condition, the hierarchical warning resulted in a significantly lower saccade count than the equivalent warning mode ($p = 0.001$) and the baseline ($p = 0.004$). Figure 6 shows the results of time to first fixation and the saccade count of different warning modes in different target number conditions.

3.2.3. Stationary gaze entropy (H_s)

There was a significant main effect of the warning mode on the H_s ($F(2, 285) = 21.067$, $p < 0.001$). Driver showed a higher H_s in the equivalent warning mode than the hierarchical warning mode ($p < 0.001$) and the baseline ($p = 0.046$),

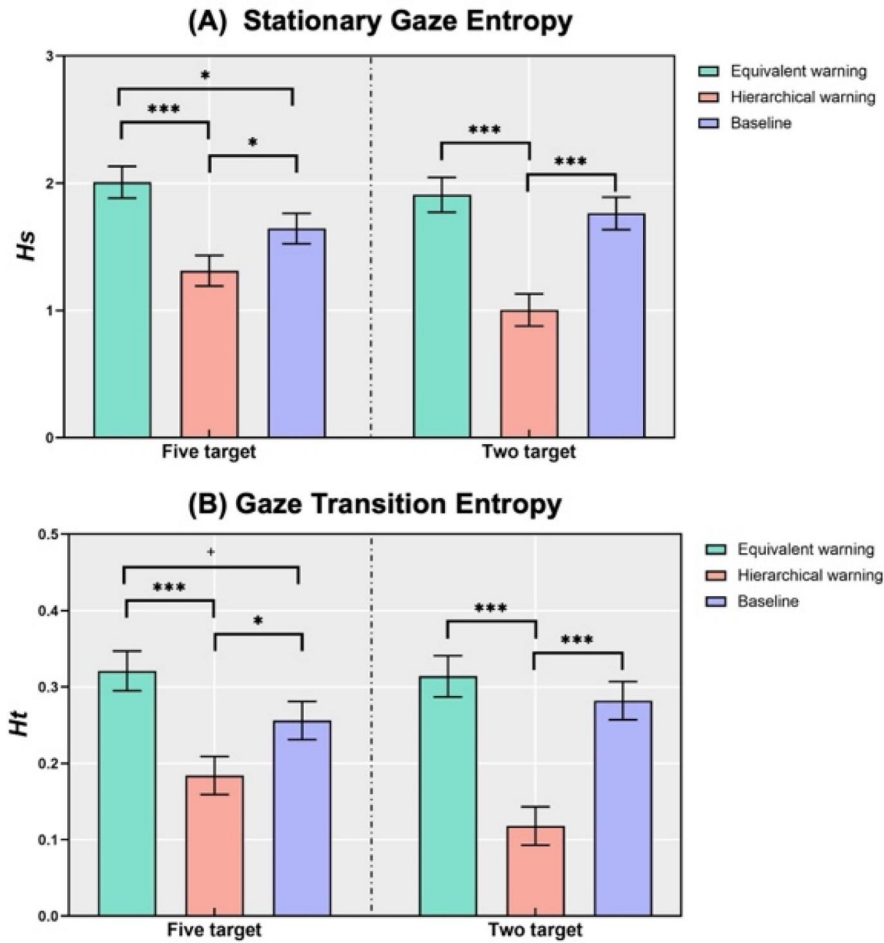


Figure 7. Effects of warning modes on the stationary gaze entropy and the gaze transition entropy in different target number conditions. Notes: $+p \leq 0.1$, $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$.

and had the lowest H_s in the hierarchical warning mode ($ps < 0.001$). The target number did not have a significant impact on drivers' H_s ($F(1, 285) = 1.637$, $p = 0.350$).

No significant interaction was reported between warning modes and the target number ($F(2, 285) = 3.547$, $p = 0.225$). We found that in the five-target condition, the equivalent warning was accompanied by the higher H_s than the hierarchical warning mode ($p < 0.001$) and the baseline ($p = 0.036$), and the hierarchical warning was accompanied by a lower H_s than the baseline ($p = 0.049$). Moreover, in the two-target condition, the hierarchical warning resulted in a lower H_s than the equivalent warning mode and the baseline ($ps < 0.001$).

3.2.4. Gaze transition entropy (H_t)

A significant main effect of the warning mode on the H_t was found ($F(2, 285) = 21.629$, $p < 0.001$). The driver showed the lowest H_t in the hierarchical warning mode ($ps < 0.001$), and in the equivalent warning mode, drivers showed marginally higher H_t than baseline ($p = 0.066$). No significant impact of target number on drivers' H_t ($F(1, 285) = 0.568$, $p = 0.452$).

No significant interaction was reported between warning modes and the target number ($F(2, 285) = 1.868$, $p = 0.156$). The result showed that in the five-target condition, drivers showed lower H_t in the hierarchical warning mode than the equivalent

warning mode ($p < 0.001$) and the baseline ($p = 0.041$), and the equivalent warning mode was associated with a marginally higher H_t than baseline ($p = 0.070$). In the two-target condition, the hierarchical warning resulted in the lowest H_t ($ps < 0.001$). Figure 7 shows the results of H_s and H_t of different warning modes in different target number conditions.

4. Discussion

In this study, we compare the impact of equivalent warning mode and hierarchical warning mode on novice drivers' reaction behavior and eye movements to verify the effectiveness of the warning priority design under multi-target augmentation conditions. We observe that the hierarchical warning mode shortens reaction times and reduces saccade counts and gaze entropy in comparison to those in the equivalent warning mode, demonstrating the superior performance of the priority design. We further analyze these results in the subsequence and summarize the implications for future AR-HUD design.

4.1. Hierarchical warning in multi-target augmentation

Firstly, our analysis of reaction times indicates that across varying target numbers, the equivalent warning mode

consistently exhibits the longest reaction times, while the hierarchical warning mode consistently alleviates this negative impact and records the shortest reaction times in all scenarios. The reaction time to a critical target is essential for driving safety (Mackenzie & Harris, 2015), and thus, this outcome preliminarily offers empirical support for the integration of warning priority design with multi-target scenarios, as a necessary and effective approach to mitigate the extended reaction times associated with multi-target augmentation, ensuring driving safety.

Upon further examination of the time to first fixation, we encountered an unexpected result. The hierarchical warning mode achieves the shortest time to first fixation under a two-target condition, yet no significant difference is observed in the first fixation time between the hierarchical warning mode and the equivalent warning mode, under a five-target condition. The time to first fixation is a critical measure of a driver's risk perception capability (Cheng et al., 2023). Consequently, the hierarchical warning mode, despite its utility, appears to have limited effectiveness in enhancing risk perception in multi-target scenarios, especially with an increased number of targets. We believe that this result once again emphasizes the strong impact of attention dispersion under multi-target augmentation, causing participants' attention to be divided among several targets, which leads to an extended time to achieve the first fixation even under warning priority design conditions. It reminds that the color-based warning priority design may not substantially improve a driver's risk perception at the perception stage and may not constitute the most effective warning form. This insight diverges from the prevailing notion that color prominence is paramount for enhancing visual search efficiency (Beck et al., 2010). However, we hypothesize that this discrepancy could also be attributed to the influence of background colors and ambient lighting on the visibility of AR graphics (Gabbard et al., 2022; Merenda et al., 2016), which might impair the effectiveness of warning priority in the perception stage. In addition, we think that another possibility is due to our use of bounding boxes for warning within the scene, where color differences are mainly reflected in the contours of the box graphics, which may, to some extent, weaken the effect of color-based warning priority.

The results presented elicit a critical question: why does the hierarchical warning mode achieve the shortest reaction times without concomitantly reducing the time to the first fixation? For instance, despite equivalent times to the first fixation, the equivalent warning mode is associated with the longest reaction times, whereas the hierarchical warning mode significantly reduces this duration. In light of these findings, we seek to elucidate their implications by integrating them with the concept of situational awareness. Situational awareness is essential for effective decision-making, particularly in dynamic settings (Endsley & Endsley, 1995). It encompasses three levels: detecting environmental elements (perception, level 1), comprehending their significance (understanding, level 2), and anticipating future developments (prediction, level 3). Consequently, the progression from the

emergence of a risky target to participant response can be divided into two stages (Cheng et al., 2023; Mackenzie & Harris, 2015), including a perception stage, which involves identifying targets and corresponds to level 1 of situational awareness, and a verification stage, which involves assessing risks and is associated with levels 2 and 3 of situational awareness. We posit that warning modes differentially affect the verification stage, thereby influencing the development of levels 2 and 3 of situational awareness in drivers. Notably, situational awareness necessitates the integration of disparate elements. For instance, level 2 transcends mere awareness of present elements to include an appreciation of their significance (Endsley & Endsley, 1995; Teichmann & Motus, 2021). Thus, it is the synthesis of detailed situational information and how this information is assembled for understanding and projection that ultimately shapes decision-making (Endsley & Endsley, 1995). During the experiment, the equivalent warning mode's prolonged reaction times are likely attributable to its absence of a warning priority design, employing similar yellow AR graphics that may obscure the distinction between risky and non-risky targets. This similarity can not only impede the identification of critical objects (Chichero & Herzog, 2015), but also impair drivers' comprehension of the meaning of the AR graphics. The ambiguity associated with this equivalent augmentation is a particular challenge for novice drivers since they are not good at recognizing potential hazards (Underwood et al., 2003). Consequently, even after fixating on the risky target, drivers must conduct additional scans to verify the risk levels of other targets, and confirm the meaning of these warning cues, prolonging the reaction time. In contrast, the hierarchical warning mode, incorporating a color-based warning priority design, conveys distinct semantic meanings through the AR graphics' colors, delineating the boundaries between the risky target and other targets. This distinction reduces the similarity between risky and remaining targets, reduces unnecessary scanning of non-risky targets, simplifies the validation process, promotes the progression of situational awareness, and leads to the shortest response time. It is especially beneficial for novice drivers. Overall, the extent to which displays provide information that is processed and integrated positively impacts situational awareness, which in turn influences the final decision (Endsley & Endsley, 1995). This finding confirms that the benefits of hierarchical warning modes extend beyond aiding in the perception of critical warnings; more importantly, they reduce the cognitive load associated with the ambiguity that arises from the concurrent presentation of AR graphics. Considering the varying importance of objects within the driving environment, we posit that the warning classification provided by the priority design is particularly indispensable under conditions characterized by multi-target augmentation.

The analysis of saccadic eye movements also supports the potential efficacy of hierarchical warning systems in mitigating superfluous scanning behaviors and promoting a higher level of situational awareness. For example, the equivalent warning mode, devoid of integrated warning priority, is observed to produce the highest saccade counts. In contrast, the hierarchical warning mode, which incorporates the

warning priority design, demonstrates a significant reduction in saccades. Elevated saccade counts are generally indicative of a more exhaustive visual search and are positively correlated with increased reaction times (Kotval & Goldberg, 1998; Zelinsky & Sheinberg, 1995). Consequently, the diminished saccade counts observed in the hierarchical warning mode substantiate the notion that integrating warning priority in scenarios with multiple targets can curtail unnecessary driver scanning, thereby abbreviating the verification phase and reaction time.

Furthermore, integrating gaze entropy measurements into our study provides a fresh vantage point for the enhanced comprehension of the influence of warning modes. Gaze entropy serves as a metric for assessing visual scanning behaviors during tasks with substantial visuospatial complexity, such as driving (Shiferaw et al., 2018). It encapsulates the intricacy of visual scanning as dictated by the statistical distribution of elementary visual features or the information density within the visual field (Shiferaw et al., 2019). Notably, stationary gaze entropy measures the overall spatial dispersion of gaze, whereas gaze transition entropy captures the complexity of gaze shift patterns, indicative of search efficiency (Shiferaw et al., 2019). Higher values of stationary gaze entropy denote a more extensive dispersion of gaze (Shiferaw et al., 2018), whereas elevated gaze transition entropy suggests increased randomness in eye movements and a higher frequency of gaze shifts among interest areas (Ebeid & Gwizdka, 2018). Consequently, this experiment employs gaze entropy to ascertain the efficacy of warning modes in modulating driver scanning behavior within hazardous driving contexts from a quantitative standpoint. Consistent with expectations, the equivalent warning mode is associated with markedly higher levels of both stationary and gaze transition entropy, signifying a wider gaze span and more frequent shifts in attention, indicative of greater visual disorder. Conversely, the hierarchical warning mode is characterized by the lowest levels of these entropy measures, implying a more focused gaze and reduced attentional shifts, thereby robustly evidencing the efficacy of warning priority design in curtailing superfluous scanning behaviors among drivers. This result confirms our suspicion to some extent.

In summary, the findings validate the hypothesis that integrating a color-based warning priority system with multi-target augmentation—referred to as the hierarchical warning mode—effectively mitigates the adverse effects typically associated with multi-target scenarios. This is demonstrated by a significant decrease in saccade count, stationary gaze entropy, and gaze transition entropy. Specifically, the efficacy of the color-based warning priority design is likely attributed to its capacity to reduce unnecessary scanning activities during the verification stage, which pertains to levels 2 and 3 of situational awareness. This design demonstrates effectiveness in minimizing additional cognitive demands stemming from the ambiguity of multiple targets and in facilitating the driver's comprehension of multiple warnings in scenarios involving multi-target augmentation.

4.2. Implications for AR-HUD design

Prior research on Augmented Reality Head-Up Display (AR-HUD) interface design has demonstrated that AR integration can facilitate driver detection of hazardous targets. This study advances the supportive role of AR to scenarios involving multiple target augmentations, offering innovative insights for the evolution of AR-HUD design. Firstly, our research corroborates the notion that employing similar AR graphics without an integrated warning priority can degrade a driver's ability to discern risky targets. Concurrently, the incorporation of warning priority into interface design emerges as an efficacious strategy to neutralize this adverse effect, enabling drivers to discern risky targets among many, expedite the verification process, and promote a higher level of situational awareness. This underscores the criticality of integrating warning priority design considerations for AR graphics in AR-HUD multi-target augmentation interfaces. Furthermore, the research analysis posits that the impact of AR graphics is predominantly evident during the verification stage subsequent to perception. This finding broadens our comprehension of AR-HUD graphic effects and highlights the necessity for interface designs to account for graphic influences on the driver's verification stage to reduce cognitive interference and augment system efficacy and safety. Ultimately, while color-based warning priority design enhances driver perception of risk under conditions of limited targets, its utility is constrained in scenarios with an abundance of targets. Hence, future AR-HUD warning priority designs must consider alternative formats for optimizing both perception and verification stages and contemplate the optimal quantity of AR-augmented targets presentations.

5. Limitations

We recognize that the findings of this study are subject to limitations regarding their generalizability and evidential force. Firstly, this study, akin to previous research (Chen et al., 2024; Wang et al., 2022), utilized video clips as stimuli with the aim of investigating the impact of multi-target augmentation on drivers' visual perception while controlling for extraneous variables through video experiments. Despite efforts to replicate real driving conditions, the interactive complexities of actual driving may exceed those observed in the video (Mackenzie & Harris, 2015), potentially undermining the ecological validity of our findings and their real-world applicability. Future research should integrate simulated driving to assess the impact of multi-target augmentation on driving performance. Second, the study's focus on novice drivers offers insights into AR-HUD interface design tailored to this demographic. However, experienced drivers possess more refined visual attention allocation and scene comprehension, suggesting their performance with AR-HUD multi-target augmentation may differ significantly. Therefore, the study's emphasis on novice drivers limits the generalizability of the results, and future studies should include experienced drivers for a more comprehensive AR-HUD interface design. Third, as noted in the introduction, real AR-HUD use involves depth perception and focus-switching challenges, leading to

accommodation-convergence conflicts (Kang et al., 2019). Multi-target augmentation may intensify these effects on visual attention. The current video experiment's format limits the exploration of multi-target augmentation effects in depth-varied scenarios, impacting the generalizability of our findings. Future studies should explore multi-target augmentation effects in more realistic settings with depth perception differences. Fourth, while our study considered scenarios with two and five targets based on previous research, real-world driving scenarios are more diverse, and individuals' multi-target perception thresholds may vary due to scenario complexity and dynamics. Future research should examine a broader range of target numbers to better understand multi-target perception processes in dynamic driving scenarios and broaden the scope of multi-target augmentation research. Notably, this study did not include a control condition with augmentation of only the risk target, an aspect that should be integrated into future research to better comprehend the impacts of multi-target augmentation in AR-HUD systems. Lastly, while the discussion section touches on the influence of AR graphics during the verification stage, the eye movement analysis conducted was not exhaustive and did not specifically dissect this stage. Subsequent studies could benefit from incorporating more detailed eye-tracking methodologies to elucidate this stage.

6. Conclusion

This research explores the effects of the AR-HUD on driver behavior within a multi-target environment, with a focus on reaction times and eye movement patterns in response to risky targets. Our goal was to evaluate the efficacy of warning priority design within multi-target scenarios. By comparing the outcomes of the equivalent warning mode, hierarchical warning mode, and a baseline group, we have advanced our understanding of the impact of multi-target conditions and the effectiveness of warning priority design within this context. The hierarchical warning mode notably exhibited superior performance, characterized by the shortest reaction times, minimal saccade count, and reduced gaze entropy across different target number conditions, thereby enhancing the efficiency of driver responses and the precision of visual scanning. Our analysis suggests that the hierarchical warning mode's effectiveness may stem from its optimization of the situational awareness in the verification stage. The results, encompassing reaction time, eye movement metrics, and gaze entropy, collectively endorse the utility of warning prioritization in AR-HUDs under conditions of multi-target augmentation. These insights not only broaden our comprehension of AR graphics' role in multi-target scenarios but also offer guidance for the future evolution of AR-HUD technology. It is important to note that the time to first fixation did not significantly vary with an increasing number of targets in the hierarchical warning mode relative to other modes. This finding implies that the color-based priority warning system's effectiveness may diminish under conditions with a higher target density. Consequently, this insight encourages further reflection on determining the optimal quantity of AR graphics and the most effective presentation methods for priority information.

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